

Monte Carlo Analysis of a BJT Memory Cell

This essay is a case study of parameter distribution and reliability in discrete devices, specifically discussing a bipolar transistor memory cell built using a pair of BCX19 NPN BJTs. The analysis was performed by a combination of LTspice simulations involving the operating point, transients, and parametric considerations such as time constants, settling times, temperature dependence, and average and dispersion values obtained from Monte Carlo analysis.

The memory cell is built from two BCX19 bipolar transistors, connected in a cross-coupled configuration. Each base is connected to the collector of the other transistor, while the collectors are connected to an active load, simulated by a resistor whose value varies with temperature ($t_c = -0.001$ for polysilicon). The active loads are connected to a power supply. The output of the memory cell is read by using the nodes of the two collectors, which in the circuit, have been labeled "Vout" and "Voutb", respectively. Both the emitters and the substrates (internally connected) have been connected to ground voltage.

For simplicity, we will consider the active loads of both transistors to be identical. The Monte Carlo analysis will be performed on a single transistor out of the two (the modified model "BCX19mc"), in order to be able to compare the behavior of the parameter-variant device with that of a device with ideal characteristics.

Feedback behavior and impedances

In a typical common emitter stage, $R_{eq} = R_o \parallel R_{load}$. But, the memory cell is a feedback circuit, and, for example, by increasing the active load of the transistor Q1, the current through its load decreases, thus restricting the base current through the transistor Q2. A reduced base current, in turn, means that a reduced current is allowed to flow through the collector of Q2. Due to the fact that the output conductance of the transistor is directly given by its Early voltage with respect to the collector current:

$$g_o = I_c / V_a = 1 / r_o$$

It results that, by decreasing the collector current of Q2, its output impedance increases. Thus, in this feedback configuration, the increase of the active load of Q1 results in an increase of Q2's output impedance, and vice versa, considering identical transistors and active loads. It is not possible to configure the circuit with an identical value for the transistors and their loads simultaneously, as increasing one increases the other, thus, they cannot converge to the same value.

Time constants

The time constants of the circuit can be obtained by observing the Bode plot obtained from the .ac analysis. However, we are only interested in simulating the low-frequency behavior of the circuit, for which a single dominant time constant influences the response time. The parameters of this time constant can be obtained from the .op analysis, specifically, the output impedances and the parasitic capacitances of the transistors. The formula for the time constant is:

$$\tau = R_{eq} * C_{eq}$$

with the equivalent resistance R_{eq} having the value, for a typical common emitter stage:

$$R_{eq} = r_o \parallel R_{load}$$

However, the cross-coupled configuration requires considerations for feedback behavior. As previously explained, the increase of the impedance of the active load creates a negative feedback which decreases the impedance of the transistor, i.e. a reduction of the equivalent resistance owed to the gain of the circuit:

$$R_{eq} = (r_o \parallel R_{load}) / (1 + A)$$

with A being the feedback loop gain, given in this case by

$$g_m * (r_o \parallel R_{load})$$

with a small R_{load} , $r_o \parallel R_{load}$ is approximately equivalent to R_{load} , and finally:

$$R_{eq} = R_{load} / (1 + g_m * R_{load})$$

For the parasitic capacitances, the equivalent value is:

$$C_{eq} = C_{be} + (1 + A) * C_{bc}$$

where C_{bc} increases the output capacitance due to the Miller effect, based on the transistor's gain.

All of these values can be obtained from the results of the .op analysis, and by introducing them into the formulas described above, we obtain:

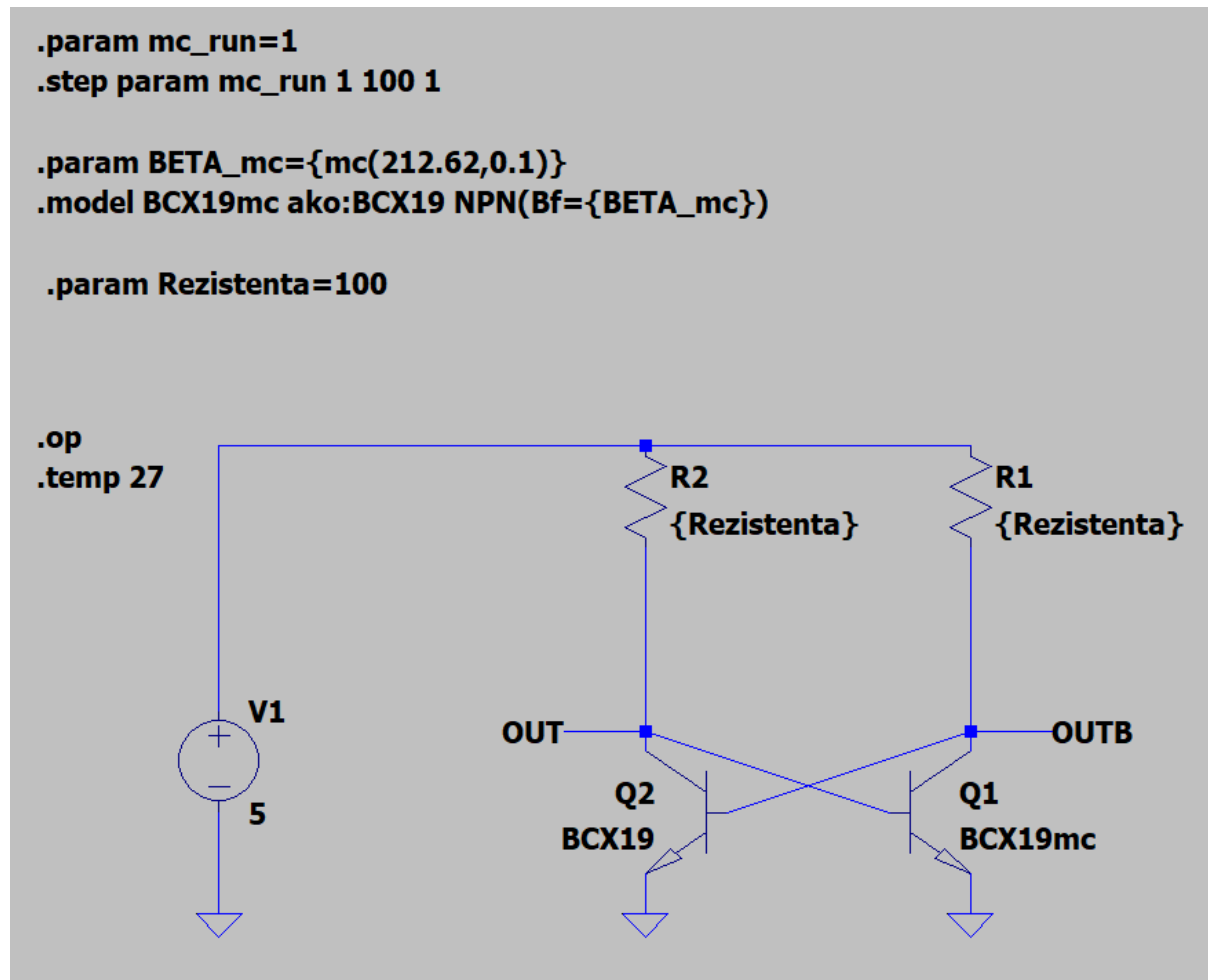
Handwritten calculations on graph paper:

$$R_{eq} = \frac{10 \Omega}{1 + 7.5 \cdot 10 \Omega \cdot \frac{1}{\Omega}} = \frac{10}{76} \Omega = 0.13 \Omega$$
$$C_{eq} = 1.8 \cdot 10^{-7} + 1.4 \cdot 10^{-11} \cdot (1 + 7.5 \cdot 10) \approx 1.8 \cdot 10^{-7} = 180 \text{ nF}$$
$$\Rightarrow \tau = 0.13 \Omega \cdot 180 \text{ nF} = 24 \text{ ns}$$

.op and .mc

The output node voltages Vout and Voutb had their operating points analyzed in typical conditions (27degC) and accounting for the Monte Carlo variance ($\pm 10\%$) of the transistor Q1's gain.

Schematic:



Netlist with temperature-variant resistors:

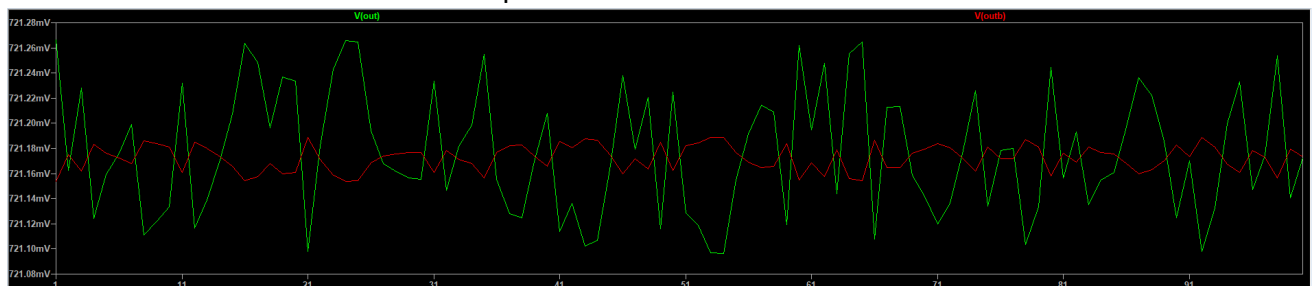
```
* C:\Users\traian\upb\rqc\1.1 op. rezistente\schematic.asc
Q1 OUTB OUT 0 0 BCX19mc
Q2 OUT OUTB 0 0 BCX19
R1 N001 OUTB {Rezistenta} tc=-0.001
R2 N001 OUT {Rezistenta} tc=-0.001
V1 N001 0 5
.model NPN NPN
.model PNP PNP
.lib C:\Users\traian\Documents\LTspiceXVII\lib\cmp\standard.bjt
.step param mc_run 1 100 1
.model BCX19mc ako:BCX19 NPN(Bf={BETA_mc})
.param BETA_mc={mc(212.62,0.1)}
.param mc_run=1
```

```
.op
.param Resistenta=100
.temp 27
.backanno
.end
```

Operating point parameters:

Name:	q2	q1
Model:	bcx19	bcx19mc
Ib:	2.30e-04	2.54e-04
Ic:	4.25e-02	4.26e-02
Vbe:	7.21e-01	7.21e-01
Vbc:	-1.13e-04	1.13e-04
Vce:	7.21e-01	7.21e-01
BetaDC:	1.85e+02	1.67e+02
Gm:	1.54e+00	1.54e+00
Rpi:	1.15e+02	1.03e+02
Rx:	3.80e+00	3.80e+00
Ro:	2.33e+03	2.33e+03
Cbe:	7.39e-10	7.39e-10
Cbc:	1.39e-11	1.39e-11
Cjs:	0.00e+00	0.00e+00
BetaAC:	1.76e+02	1.59e+02
Cbx:	0.00e+00	0.00e+00
Ft:	3.25e+08	3.25e+08

Parametric variations of the .MC sweeps:



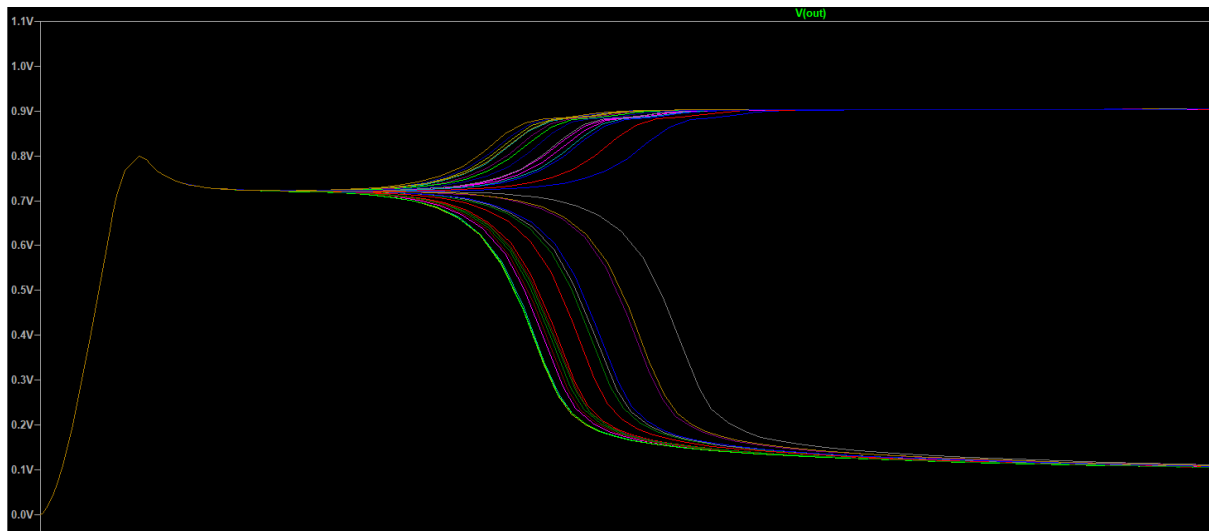
The results were tabulated, and their average and standard deviation were computed:

	Vout -- Q2 (ideal)	Voutb -- Q1 (mc)
Average	7,21E-01	7,21E-01
Dispersion	1,01E-05	4,89E-05

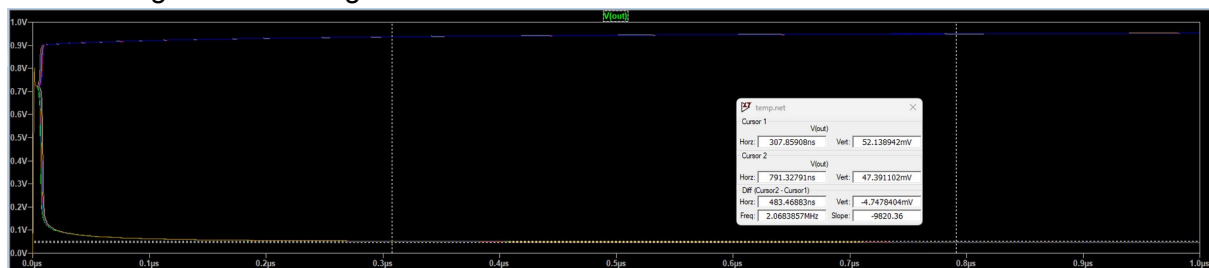
.tran and .mc

A pulse voltage source with 5V matching the source used in the .op small signal analysis, from which the parasitic capacitances were deduced, will be used to simulate the settling behavior at 27 degrees celsius.

Settling waveform:



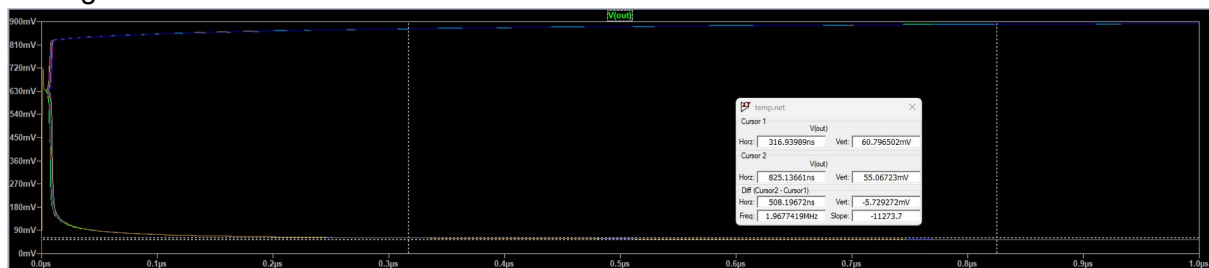
90% settling time at 27 degrees:



Similar plots are obtained for:
0 degrees Celsius:



85 degrees Celsius:



Settling time:

Minimum: 284ns

Typical: 308ns
Maximum: 317ns

Conclusions

Parametric variations in bipolar junction transistors introduce significant deviations in their resulting behavior, with the device's metastable state being able to settle into either of two possibilities ($\sim 0V$ or $\sim 1V$ for this circuit) as a consequence of the $\pm 10\%$ variation of the transistor's gain. This reflects the low fault tolerance of the BJT when it comes to manufacturing, as the functionality of each device is significantly altered by its particular imperfections.